Infosys — Aptitude Test 3

Questions 51–75 of 100 · 25 questions

Question 51. When a train travels at a speed of 60kmph, it reaches the destination on time. When the same train travels at a speed of 50kmph,it reaches its destination 15min late. What is the length of journey?

Solution:
St = speed of train; Sc = speed of car
8 = (120/St) + (480/Sc) [Time=D/S]
8.33 = (200/St) + (400/Sc)

Solution:
Assume the distance of journey = D
Actual time taken is ‘T’ at actual speed ‘S’.
Time = Distance/Speed
Time taken in first case;
T = D/60
---(1)
Time taken in second case;
T + (1/4) = D/50
T = (D/50) – (1/4)
T = (4D-50)/200 ---(2)
From (1) and (2)
D/60 = (4D – 50)/200
By solving we get, D = 75km.

120
480
St
Sc
+

8

=

Question 52. A participated in cycling contest and he drove the lap at the rate of 6kmph, 12kmph, 18kmph, 24kmph. What is his average speed?

Solution:

4 =

A.S

1

1
+

1
+
■A.S=11.52

1
+

6

12

18

24

Question 53. In a journey of 600 km, due to some problem in the vehicle, speed was reduced by 200 kmph and it takes 30min extra, Find the Actual time taken for journey?

Solution:
Let T be the actual time and S be actual speed
Time = Distance/Speed
Time taken in first case;

T = 600/S
---(1)
Time taken in second case;
T + (1/2) = 600/(S – 200)
T = [600/(S – 200)] – (1/2)
T = (1400 - S)/(2S – 400)
---(2)
From (1) and (2)
600/S = (1400-S)/(2S-400)
1200S – 240000 = 1400S – S2

S2 - 200S – 240000 = 0, by solving we get,
S = 600kmph.
T = 600/S = 600/600 = 1 hour.

Question 54. In a 200m race, if A beats B by 10 meters and B beats C by 15 meters then A beats C by how many meters?

Solution:
When A runs 200m, B runs 190m
When B runs 200, C runs 185m
Therefore, 200 : 185 = 190 : x, where x is

the
distance
run
by
C
when
A
completes the race
x = 190(185)/200
x = 175.75m
A beats C by 200 – 175.75
= 24.25 meters

Question 55. A girl goes to her office for work which is 50 miles. She goes to her office few distance by bicycle and remaining by train. The speed of bicycle is 15 mph and that of train is twice of bicycle. If she spend 20 min. more on bicycle, then total time taken by her from going to office from her home?

Solution:
Speed of cycle = 15mph

Speed of train
= 2 x 15
= 30mph
From the question, if the girl travels 1 hour in both cycle and train, the distance covered is

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= 15 + 30
= 45
If the girl travels 20 minutes extra by cycle,
The distance travelled = 15x(20/60)

= 5 miles
Total distance
= 45 + 5 = 50 miles
Total time taken,

= 1hr + 1hr + 20 min
= 2hr 20 min

Question 56. A car travelling with 5/7 of its actual speed covers 42 km in 1 hour 40 min 48 sec. Find the actual speed of the car.

Solution:
Speed of the car to travel 42 km in 1:40:48
= 42km/(1:40:48)hour
= 42km/(3600+2400+48 sec)
= 42/6048 km per sec

6048sec
= 6048/3600 hours
= 1.68 hours
Speed
= 42/1.68
= 25kmph
25kmph = (5/7)Actual speed.
Actual speed = 25(7/5) = 35kmph

Question 57. A man completes a journey in 10 hours. He travels first half of the journey at the rate of 21 km/hr and second half at the rate of 24 km/hr. Find the total journey distance in km.

Solution:
Let the total distance be 2D.
He travels distance D at a speed of
21kmph
The remaining distance D at 24kmph
speed
Total time taken:
Time for first half + time for second half
= (D/21) + (D/24) = 10
24D + 21D = 10x504
45D = 5040 ■D = 112 km

Question 58. If 1st day of a month is Thursday, then find the number of days in that month if the last day of month is 5th

time and wrist watch.

I n f o s y s

Saturday .

Solution:
If a month starts on Thursday, 29th day
of that month will be Thursday, 30th day
will be Friday and 31st day will be
Saturday.
The month mentioned in the question
has 31 days.

A : E = Actual time : wrist watch time
6 : E = 5184 : 5005
E = 6x5005/5184
E = 5 hours 47 min 34 seconds

Question 59. The quarter of the time from

midnight to present time added to the half of the time from the present to midnight gives the present time. What is the present time?

Question 60. When the actual time pass 1 hour, the wall clock is 10 min behind it. When 1 hour is shown by wall clock, the table clock shows 10 min ahead of it. When the table clock shows 1 hour, the alarm clock goes 5 min behind it. When alarm clock goes 1 hour, the wrist watch is 5 min ahead of it. Assuming that all clocks are correct with actual time at 12 noon what will be time shown by wrist watch after 6 hours?

Solution:
Ratio between actual time to wall clock
= 60 : 50 = 6 : 5 = A : B
Ratio between wall clock and table clock
= 60 : 70 = 6 : 7 = B : C
Ratio between table clock and alarm
= 60 : 55 = 12 : 11 = C : D
Ratio between alarm and wrist watch
= 60 : 65 = 12 : 13 = D : E
We have to find the ratio between actual

Question 61. Sometime after 10:00 PM a murder took place. A witness claimed that the clock must have stopped at the time of the shooting. It was later found that the position of both the hands were the same but their positions had interchanged. Tell the time of the shooting (both actual and claimed).

Solution:
Let the present time be T hours.
Time from now to midnight = 24 – T
Given, (1/4)T + (1/2)(24 – T) = T
T/4 + (48 – 2T)/4 = T
T + 48 – 2T = 4T

5T = 48
T = 9.6 hours = 9 hours 36 minutes

Solution:
Let the claimed time be H1 hours M1 minutes
The actual time of death is H2 hours M2 mins
Since the position of the hands are swapped, M1 is H2 and H1 is M2
Angle of M2 = Angle of H1
6M2 = 30H1 + M1/2
---(1)
30H1 ■Angle of hour hand at H1 hours
M1/2 ■Extra angle of hour hand for M1 mins
Angle of M1 = Angle of H2
6M1 = 30H2 + M2/2
---(2)
(1)+(2) ■11(M1 + M2) = 60(H1 + H2)
---(3)
(1)–(2) ■13(M1 – M2) = 60(H2 – H1)
---(4)
By substituting values H1 = 10; H2 = 11
Claimed time = 10:59
Actual time = 11:54

Question 62. A clock showing 6’o’clock takes 30 seconds to strike 6 times. How long will it take to strike 12 at midnight?

Solution:

A

A

C
x

D
x
=

B
x

E

B

C

D

E

5184

6

12
x

12
x
=

6
x

5005

5

7

11

13

The first bell will ring at zero seconds.
So, at 6’o’clock the remaining 5 bells will ring in 30 seconds at an interval of 6 seconds per bell.
The same case happens at 12’o’clock.
The first bell will ring at zero seconds.
The remaining 11 bells will take
11 x 6 = 66 seconds

Question 63. Find the time at which minute and hour hands are at same position between 9am and 10am.

Solution:
Use the following formula to find out the time at which two hands will be overlapping.
60H/11
Where H = least hour among the two
Here H = 9
60x9/11 = 49.09
= 49 mins 6 seconds
They will overlap @ 09:49:06 AM

Question 64. A cube is divided into 729 identical cubelets. Each cut is made parallel to some surface of the cube . But before doing that the cube is colored with green color on one set of adjacent faces ,red on the other set of adjacent faces, blue on the third set. So, how many cubelets are

there which are painted with exactly one color?

Solution:
Formula for number of cubes painted on only one face:
= 6(n1/3-2)2

Where n = number of small identical cubes.
Number of cubes painted on one face
= 6(7291/3 – 2)2

= 6(9-2)2

= 6(49) = 294

Question 65. We need to carve out 125 identical cubes from a cube . what is the minimum number of cuts needed?

Solution:

Formula:
Minimum number of cuts = 3(n1/3 – 1)
Where n is the number of small identical cubes
Minimum number of cuts
= 3(1251/3 – 1)
= 3(5-1) = 12

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Question 66. A big cube painted red on all the sides. It was cut into 27 smaller cubes by 6 straight lines. How many of the smaller cubes painted on all 3 sides, on 2 sides, on 1 side and no faces painted?

Solution:
Number of small cubes painted 3 sides
= 8
Constant for any number of small cubes because, only the corners will have three faces painted and a cube has 8 corners.

Number of cubes painted two sides:
These are the cubes which are at the edges.
Formula: 2 sides = 12(n1/3 – 2)
Where n = number of small identical cubes
2 sides painted cubes = 12(271/3 – 2)
= 12

Number of cubes painted one side:
Formula: 1 sided = 12(n1/3 – 2)2

Where n = number of small identical cubes

1 sided cubes = 6(271/3 – 2)2 = 6

Number of cubes painted on no face:
Formula: No face painted = (n – 2)3 = 1

Question 67. There is a 4 inch cube painted on all sides. This is cut into no of 1 inch cubes. What is the no of cubes which have no painted sides.

Solution:
The number of small cubes obtained when a cube with side X units is cut into small cubes of side Y units is:
Number of small cubes, n = (X/Y)3

The number of small cubes obtained

= (4/1)3 = 64
Number of cubes with no faces painted
= (n – 2)3 = (4 – 2)3 = 8

Quantity of syrup would be 5 liters
From this solution 1 liter of syrup should be replaced with water.
If 1/5th of the solution is removed, 1 liter syrup and 0.6 liters water will be gone.
If we replace that with 1.6 liters of water, we will now have 4 liters of water and 4 liters of syrup.
Fraction of solution to be removed and replaced is: 1/5

I n f o s y s

Question 68. A grocer has a sales of Euro 6435, Euro 6927, Euro 6855, Euro 7230 and Euro 6562 for 5 consecutive months. How much sale must he have in the sixth month so that he gets an average sale of Euro 6500?

Solution:
The expected average for six months
= 6500
Total income till 5 months
= 6435+6927+6855+7230+6562
= 34009
Average income on sixth month= 6500
6500 = (34009 + X)/6
Income on sixth month should be
= 4991

Question 69. A person went to a shop and asked for change for 1.15paise. He insisted on getting the change in the following denominations;

50p,25p,10p and 5p. How many coins did he get in each denomination?

Solution:
It has to be solved by trial and error method or using the options.
Here it is obvious that if the person gets
1 nos of 50p coin
2 nos of 25p coin
1 nos of 10p coin and
1 nos of 5p coin
He will get a total of 1.15 Rupees

Ratio between two varieties= 1 : 3
Splitting 40 in the ratio
1 : 3, we get 10 and 40
10 pounds of Brand A was taken.

Question 70. A vessel is filled with liquid, 3 parts of which are water and 5 parts syrup. How much of the mixture must be drawn off and replaced with water so that the mixture may be half water and half syrup?

Solution:
The 7 persons should be arranged in the following pattern so that the men will not sit together.

Solution:
Let us assume that there is an 8 liter solution
Quantity of water would be 3 liters

M W M W M W M
Since the men are arranged according to their age order, they can be seated in

Using
Allegation rule

3
5

Question 71. A coffee seller has two types of coffee Brand A costing 5 bits per pound and Brand B costing 3 bits per pound. He mixes two brands to get a 40 pound mixture. He sold this at 6 bits per pound. The seller gets a profit of 33 1/2 percent. How much he has used Brand A in the mixture?

Solution:
Selling price = SP; Cost price = CP
SP = (133.33/100)CP
6 = (133.33/100)CP
CP = 4.5

Question 72. In how many ways can 4 men and 3 women can be arranged so that each men should not sit together and they must be in the order of their age?

Question 73. In how many ways can the letters in mmmnnnppqq can be arranged with two n's together?

Solution:
Arranging with 2 n's together:
Select 2 n's out of three = 3C2 = 3
The selected 2 n's are considered as one.
There
will
be
9
elements
after
combining.
Number of arrangements,
= [9!/(3! x 2! x 2!)] x (2!/2!) x 3
= 11340
In the above arrangements, some them will have 3 n's together. We have to eliminate them.
Number of arrangements in which 3 n's are together, = [8!/(3! x 2! x 2!)](3!3!)
= 420
Number of arrangements in which 2 n's are together,
= 11340 – 420 = 10920

Question 74. How many three digit numbers can be formed using 2,3,4 and 5 with none of the digits being repeated?

Solution:
nPr= 4P3 = 4 x 3 x 2 = 24

Question 75. There are 3 types of apples in a box. What is the number of apples one should take so that we end up with 3 apples of one kind. There are three apples in each kind

Solution:
There are three apples in each type.
One should take at least 4 apples to make sure to get 2 of same kind.

Similarly one must take 7 apples to get three apples of same kind.
The number of apples required to have 3 apples of different kind = 7